1

(a) 158H = 0000 1 0101 1000 0 10

GDT RPL

A C 2

= AC2H (answer).

(b) Starting = D6590B00

G=0; End = start + limit

= D6590B00 + FB000

= D668BB00H

(c) physical address = start + offset

= D6590B00 + D096E

= D666146E (answer).

(d) Access = B2H ⇒ 10110010

A=0; Not accessed. (Ans.)

(e) E=0; ED=0 ⇒ DS

(f) DPL = 01.

2. a) $\frac{\text{Resolution of the ADC}}{\text{System Voltage}} = \frac{\text{ADC Reading}}{\text{A.V.M.}}$

$\Rightarrow$ ADC Reading $= \frac{\text{R. of the ADC} \times \text{A.V.M.}}{\text{System Voltage}}$

For Device A,

$$\text{ADC Reading} = \frac{(2^8 - 1) \times 8}{10}$$

For Device B,

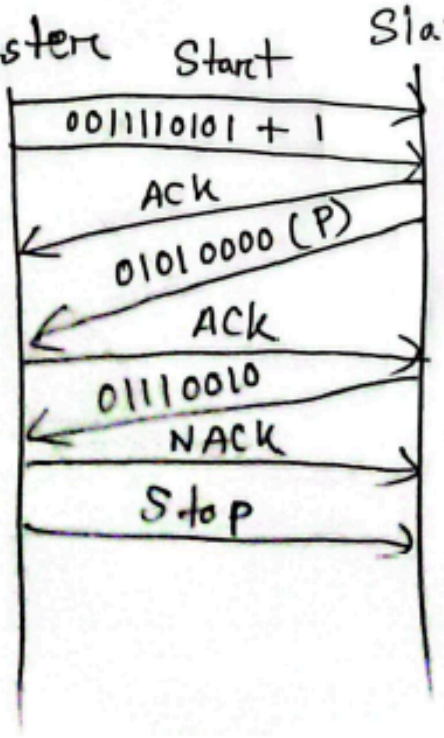
$$\text{ADC Reading} = \frac{(2^{16} - 1) \times 0.4}{x}$$

ATQ,

$$\frac{255 \times 8}{10} = \frac{65535 \times 0.4}{x}$$

$\therefore x = 128.5\ V$

b)



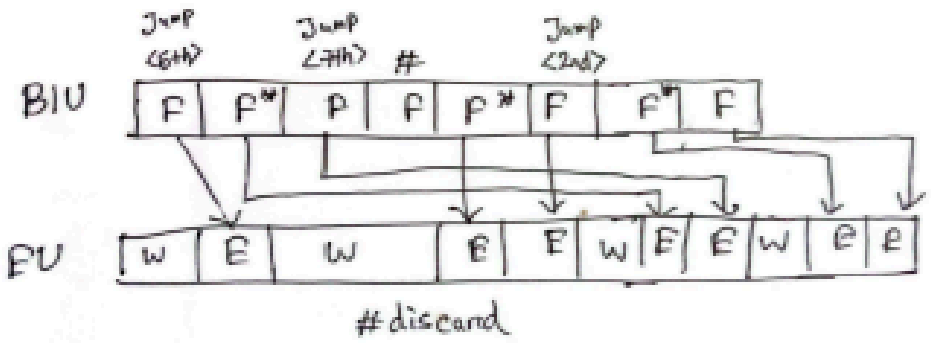
'P' = 80

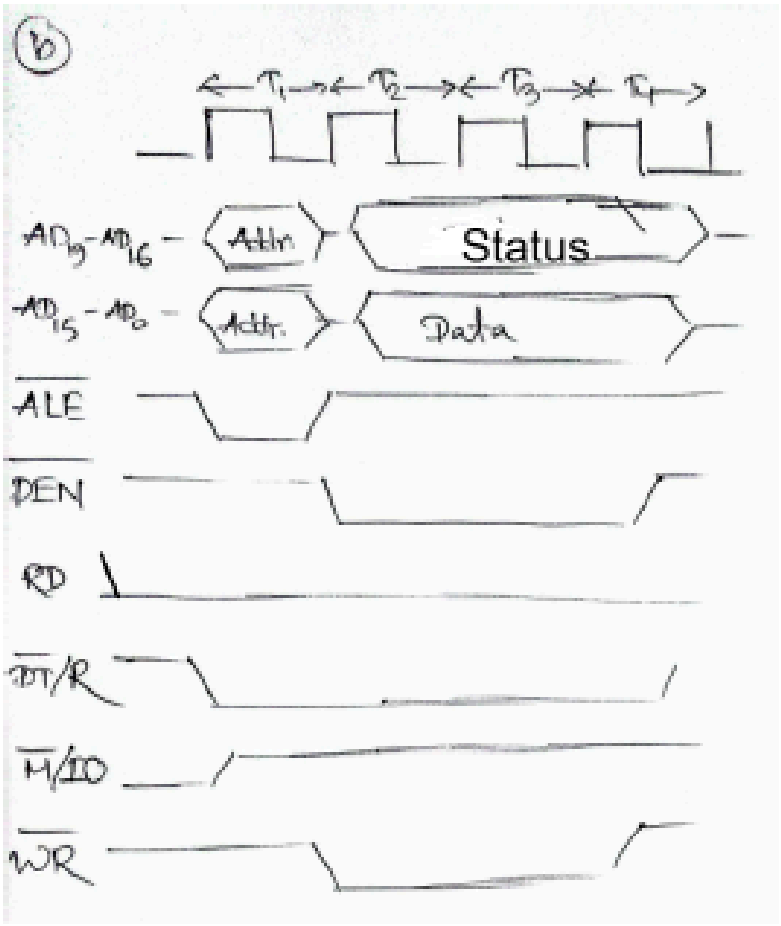
'n' = 114

c) CISC because

i. smaller number of registers

ii. Multiple tasks in one instruction





(c) #entries in the page table = $\frac{1KB}{2B}$ = 512

These reside in the primary memory.

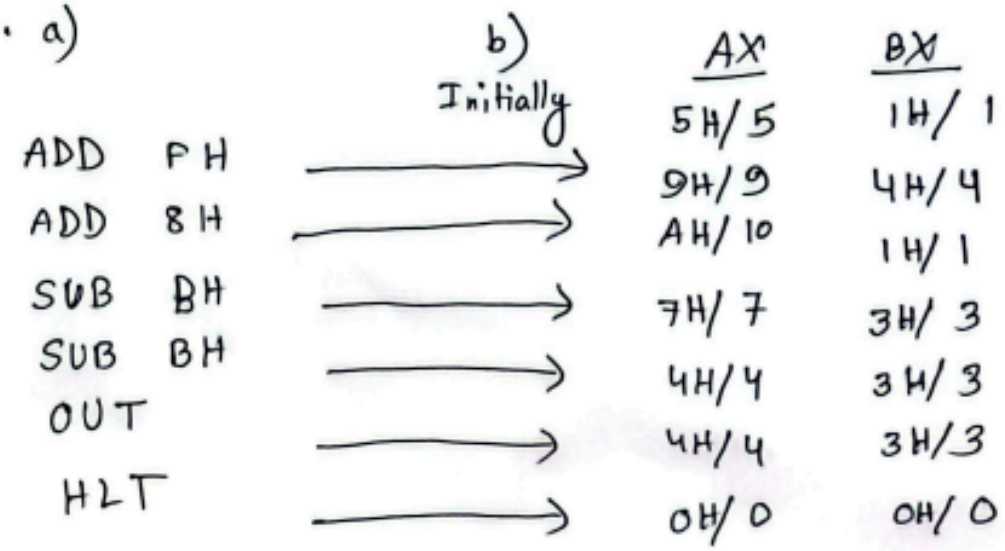
So, # pages in secondary memory

= 2048 − 512

= 1536

4. a)

b)



c) T_4,

CON = 1010 0000 0111 (A07) ($\overline{E_I}$, $\overline{L_M}$)

So, Ins. Reg → Bus

Bus → MAR

T_5,

CON = 1110 0011 0110 (E36) (CE, $\overline{L_B}$)

So, RAM → Bus

Bus → B Reg.

T_6,

CON = 1100 0010 1111 (C2F) ($\overline{L_A}$, E_U, $\overline{S_0}$)

So, Add

Adder/Subtraction → Bus

Bus → AX

So, ADD operation is being executed.